Esercizio 1. (Martina Apicella) Let $\varphi(x;z) \in L$ and $a \in \mathcal{U}^z$. Prove that the following are equivalent

1. the type $\{\varphi(x; f\,a) : f \in \mathrm{Aut}(\mathcal{U})\}$ is realized (in $\mathcal{U}$)
2. there is a formula $\psi(z) \in L$ such that $\psi(a)$ and $\exists x \forall z\,[\psi(z) \to \varphi(x;z)]$.

Esercizio 2. (Beatrice Calderaro) Let L be a countable language containing the language of groups and assume $\mathcal{U}$ is a group. Let $p(x) \subseteq L$ define a subgroup of $\mathcal{U}$. Prove that there are some formulas $\varphi_n(x) \in L$ such that

1. $p(x) \leftrightarrow \{\varphi_n(x) : n < \omega\}$
2. $\varphi_{n+1}(x) \wedge \varphi_{n+1}(y) \to \varphi_n(x \cdot y)$.

Esercizio 3. (Francesco Dam) Let L be a countable language containing the language of groups and assume $\mathcal{U}$ is a group. Prove that the following are equivalent

1. every model M is a normal subgroup of $\mathcal{U}$
2. some ω-saturated model M is a normal subgroup of $\mathcal{U}$
3. $\mathcal{U}$ is a BFC group.

A group G is BFC (has boundedly finite conjugacy classes) if for some n the sets $\{g\,a\,g^{-1} : g \in G\}$, as a ranges over G, have at most n elements.

Esercizio 4. (Filippo Giugie) Let $A \subseteq B$ and $p(x) \subseteq L(A)$. Suppose that $\mathrm{tp}(a/B)$ is isolated over B for every $a \models p(x)$. Prove that $p(x)$ is isolated over A.

Esercizio 5. (Mattia Lo Mele) Let $\mathcal{D} \subseteq \mathcal{U}$ be invariant over A. We say that $\mathcal{D}$ is quasi-definable from below if for every $a \in \mathcal{D}$ and every $\varphi(x) \in \mathrm{tp}(a/A)$ there is formula $\psi(x) \in L(A)$ such that $\psi(x) \to \varphi(x)$ and $\psi(\mathcal{U}^x) \subseteq \mathcal{D}$. We say that $\mathcal{D}$ is quasi-definable from above if $\neg\mathcal{D}$ is quasi-definable from below. Finally $\mathcal{D}$ is quasi-definable if it is both quasi-definable from below and from above.

Let $L(A)$ be countable. Prove that if $\mathcal{D}$ is quasi-definable then there is a type-definable set $\mathcal{C}$ and a model $M \supseteq A$ such that $M \cap \mathcal{C} = M \cap \mathcal{D}$.

Esercizio 6. (Andrea Morello) Let $p(x) \subseteq L(B)$ and $p_n(x) \subseteq L(A)$, for $n < \omega$, be such that

$$p(x) \to \bigvee_{i<\omega} p_i(x)$$

Prove that

$$p(x) \wedge \varphi(x) \to p_n(x)$$

for some $n < \omega$ and some formula $\varphi(x) \in L(A)$ consistent with $p(x)$.

Esercizio 7. (Gabriele Maiolino) Let x be a finite tuple. A set $\mathcal{B} \subseteq \mathcal{U}^x$ is *nowhere dense* in the topology induced by A if for every consistent formula $\varphi(x) \in L(A)$ there is some consistent $\psi(x) \in L(A)$ such that $\psi(x) \to \varphi(x)$ and $\psi(\mathcal{B}) = \varnothing$. A set $\mathcal{C} \subseteq \mathcal{U}^x$ is *meager* if it is union of countably many nowhere dense sets.

Assume $L(A)$ is countable. Prove that for every countable model M the following are equivalent

1. $\mathcal{C}$ is meager in the topology topology induced by A
2. there is a model M disjoint of $\mathcal{C}$